

Unit I

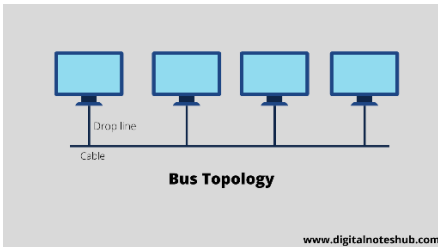
INTRODUCTION and PHYSICAL LAYER

Course objective: To learn the basic concepts of layered approach in Computer Networks and different transmission media of physical layer.

Course Outcome: Evaluate the functions and performance of a network

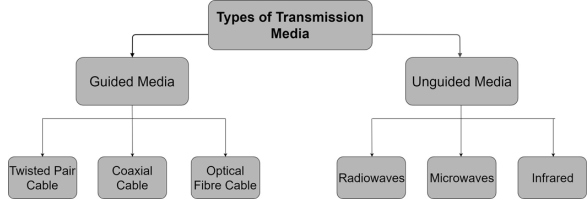
Bloom's Taxonomy Level: K1, K2, K3, K4

PART A

S.NO	Question	BTL
1	State the meaning of a computer network and list its network criteria. A network is a set of devices (often referred to as nodes) connected by communication link. A node can be a computer, printer, or any other device capable of sending or receiving data generated by other nodes on the network.	K1
2	Differentiate between LAN, MAN and WAN based on their geographical coverage. One of the major differences is the geographical area they cover, i.e. LAN covers the smallest area; MAN covers an area larger than LAN and WAN comprises the largest of all. LAN or Local Area Network connects network devices in such a way that personal computer and workstations can share data, tools and programs.	K4
3	Identify the different types of network topology. A network topology is the arrangement of nodes -- usually switches, routers, or software switch/router features -- and connections in a network, often represented as a graph. Types of topology: • Star topology, • Bus topology, • Ring topology • Mesh topology	K1
4	State the meaning of Bus Topology. Bus topology is a network type where all devices are connected to a common cable or backbone.  www.digitalnoteshub.com	K1

5	Differentiate between guided media and unguided media and list their major types. Guided Media: It is also referred to as Wired or Bounded transmission media. (i) Twisted Pair Cable (ii) Coaxial Cable (iii) Optical Fibre Cable Unguided media: It is also referred to as Wireless transmission media i) Radiowaves (ii) Microwaves (iii) Infrared	K2												
6	List the seven layers of the OSI model in their proper order. The Open Systems Interconnection model (OSI model) is a conceptual model that characterizes and standardizes the communication functions of a telecommunication or computing system. (i) Application Layer (ii) Presentation Layer (iii) Session Layer (iv) Transport Layer (v) Network Layer (vi) Datalink Layer (vii) Physical Layer	K1												
7	Differentiate TCP and UDP. <table><thead><tr><th>TCP</th><th>UDP</th></tr></thead><tbody><tr><td>Reliable</td><td>Unreliable</td></tr><tr><td>Connection-oriented</td><td>Connectionless</td></tr><tr><td>Segment retransmission and flow control through windowing</td><td>No windowing or retransmission</td></tr><tr><td>Segment sequencing</td><td>No sequencing</td></tr><tr><td>Acknowledge segments</td><td>No acknowledgement</td></tr></tbody></table>	TCP	UDP	Reliable	Unreliable	Connection-oriented	Connectionless	Segment retransmission and flow control through windowing	No windowing or retransmission	Segment sequencing	No sequencing	Acknowledge segments	No acknowledgement	K2
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8	State the components that contribute to network latency. The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. Latency is made up of four components: Latency = transmission time + queueing time + processing delay	K1												
9	Identify the factors that determine transmission time. The <i>transmission time</i>, is the amount of <i>time</i> from the beginning until the end of a message <i>transmission</i>. The time required for the transmission of a message depends on the size of the message and the bandwidth of the channel. Transmission time= message size/bandwidth.	K1												

10	List the layers of the TCP/IP protocol suite. The suite gets its name from two of the protocols that belong to it: the Transmission Control Protocol (TCP) and the Internet Protocol (IP). TCP/IP is the traditional name for this protocol suite. The TCP/IP model is a condensed version of the OSI reference model consisting of the following 5 layers: • Application Layer. • Transport Layer. • Internet Layer. • Data link layer • Physical layer	K1
11	Recall the formula used to calculate propagation time. Propagation time is the time required for a bit to travel from the source to the destination. The propagation time is calculated by dividing the distance by the propagation speed. Propagation time = distance/propagation speed.	K1
12	Discuss the purpose of a switch? A network switch (also called switching hub, bridging hub) is a computer networking device that connects devices on a computer network by using packet switching to receive, process, and forward data to the destination device.	K2
13	What is RTT? Round-trip time (RTT) is the duration in milliseconds (ms) it takes for a network request to go from a starting point to a destination and back again to the starting point.	K1
14	Mention the primary functions of TCP and IP in network communication. The difference is that TCP is responsible for the data delivery of a packet and IP is responsible for the logical addressing. In other words, IP obtains the address and TCP guarantees delivery of data to that address. ... Transmission Control Protocol (also known as TCP) is a core protocol of the Internet Protocol Suite.	K1
15	Enumerate the major functions of the Transport Layer in the OSI model. The basic function of the Transport layer is to accept data from the layer above, split it up into smaller units, pass these data units to the Network layer, and ensure that all the pieces arrive correctly at the other end. Functions of Transport Layer • Service Point Addressing • Segmentation and Reassembling: • Connection Control: • Flow Control • Error Control	K1

16	Indicate the purpose of segmentation at the Transport Layer. A message is divided into segments; each segment contains sequence number, which enables this layer in reassembling the message	K1
17	Recall the major functions of the Network Layer in the OSI model. This layer is responsible for the delivery of packets from source to destination. It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors. The other responsibilities of this layer are ☐ Logical addressing and Routing	K1
18	Specify the purpose of multiplexing in data communication. Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.	K1
19	Illustrate the types of transmission media.  <pre> graph TD A[Types of Transmission Media] --> B[Guided Media] A --> C[Unguided Media] B --> D[Twisted Pair Cable] B --> E[Coaxial Cable] B --> F[Optical Fibre Cable] C --> G[Radiowaves] C --> H[Microwaves] C --> I[Infrared] </pre>	K2
20	Name the categories of unguided transmission media. An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as wireless transmission. Unguided transmission is broadly classified into three categories : (i) Radio Waves (ii) Microwaves (iii) Infrared	K1

PART B

Q.No	Question	BTL
1	Outline the OSI model and elaborate on the functions of each layer with a neat diagram.	K2
2	Examine the different telecommunication and cellular networks along with their characteristics.	K2
3	Elaborate on guided transmission media and discuss their different categories with suitable examples. (8 Marks)	K2
4	Illustrate unguided transmission media and their different categories with suitable examples. (8 Marks)	K2
5.	Illustrate the TCP/IP protocol suite and its layers with a neat diagram.	K1
6.	Analyze the factors that influence the performance of the Physical Layer.	K4